

- 1) *ortho*-chlorocyclopropylbenzene
- 2) *ortho*-cyclopropylchlorobenzene
- 3) chlorophenylcyclopropane
- 4) cyclopropylchlorobenzene

<S> Y

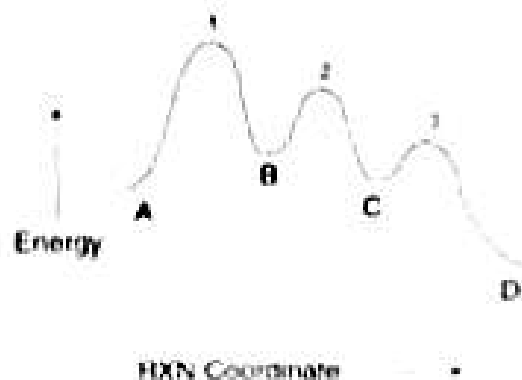
<C+> 1

<C> 2

<C> 3

<C> 4

<Q> in the energy diagram shown below, which step is the fastest?



<S> Y

<C> A to B

<C> A to 1

<C> C to 3

<C+> C to D

2) protonation of the double bond to form a carbocation.

3) removal of proton/regeneration of the acid catalyst.

4) carbocation rearrangement.

<S> Y

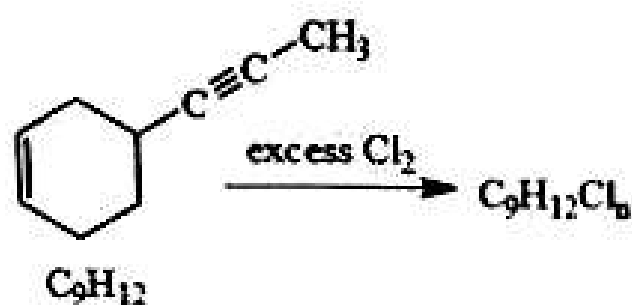
<S> 1

<C+> 2

<C> 3

<C> 4

<Q> For the reaction shown below, what is the value of n in the product?



<S> Y

<C> 2

<C> 4

<C+> 6

<C> cannot be predicted

<Q> Give an acceptable name for the following compound?

<Q> Consider the reaction ($A \rightarrow B$) that goes to completion. Which of the following statements is True?

- 1) ΔH° has a negative value and the equilibrium constant K is greater than 1.
- 2) ΔH° has a negative value and the equilibrium constant K is less than 1.
- 3) ΔH° has a positive value and the equilibrium constant K is greater than 1.
- 4) ΔH° has a positive value and the equilibrium constant K is less than 1.

<S> Y

<C+> 1

<C> 2

<C> 3

<C> 4

<Q> In the acid-catalyzed addition of water (H_2O) to propene, the rate-determining step is:

- 1) attack of water on the carbocation intermediate.
- 2) protonation of the double bond to form a carbocation.
- 3) removal of proton/regeneration of the acid catalyst.
- 4) carbocation rearrangement.

<S> Y

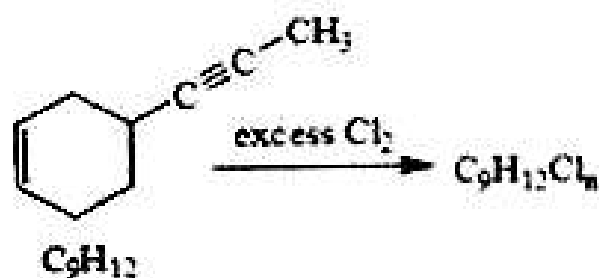
<C> 1

<C+> 2

<C> 3

<C> 4

<Q> For the reaction shown below, what is the value of n in the product?



<Q> The indicated bond is:



- 1) electrophilic because it is electron-deficient
- 2) electrophilic because it is electron-rich
- 3) nucleophilic because it is electron-deficient
- 4) nucleophilic because it is electron-rich

<S> Y

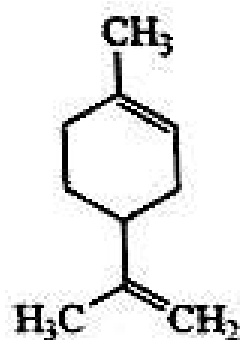
<C> 1

<C> 2

<C> 3

<C+> 4

<Q> How many different products are obtained upon ozonolysis of the following compound



<S> 1

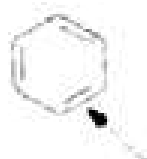
<C+> 2

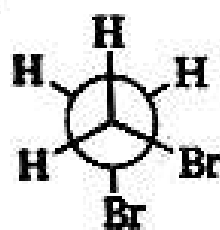
<C> 3

<C> 1

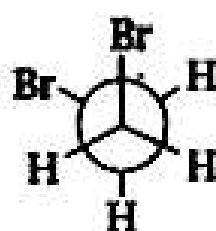
<C> 4

<Q> The indicated bond is

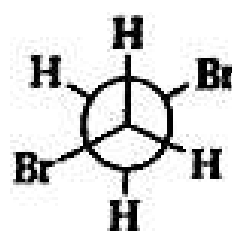




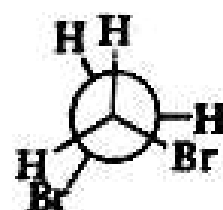
I)



II)



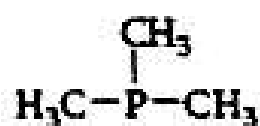
III)



IV)

- ☐ Y
☐ I
☐ II
☐ III
☐ IV

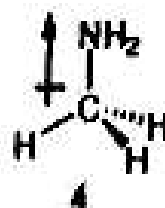
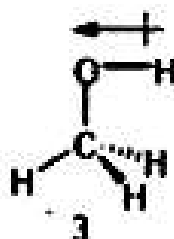
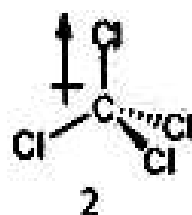
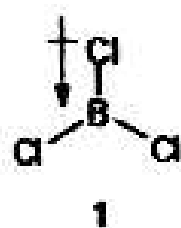
<Q> The formal charge on phosphorus (P, group 5A) in the following molecule is:



- I) 0 II) -1 III) +1 IV) +2

- ☐ Y
☐ I
☐ II
☐ III
☐ IV

<Q> Which of the following drawings of the direction of the indicated bond polarity is incorrect?



- <S> Y
- <C> 4
- <C> 3
- <C> 2
- <C+> 1

<Q> How many polar covalent bonds are present in the compound shown below?

I) 5

II) 6

III) 7

IV) 3



- <S> Y
- <C> I
- <C+> II
- <C> III
- <C> IV

<Q> Including *cis-trans* isomers, how many possible isomers are present for dimethylcyclohexane?

- <S> Y
- <C> 3
- <C> 4
- <C> 6
- <C+> 7

<Q> Which of the following interactions costs the less amount of energy?

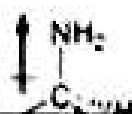
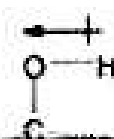
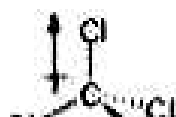
- <S> Y
- <C> CH_3 , CH_3 gauche
- <C+> CH_3 , F gauche
- <C> CH_3 , H eclipsed
- <C> CH_3 , CH_3 eclipsed

<Q> In Lewis structure of C_2H_2 , how the C-H bond formed?

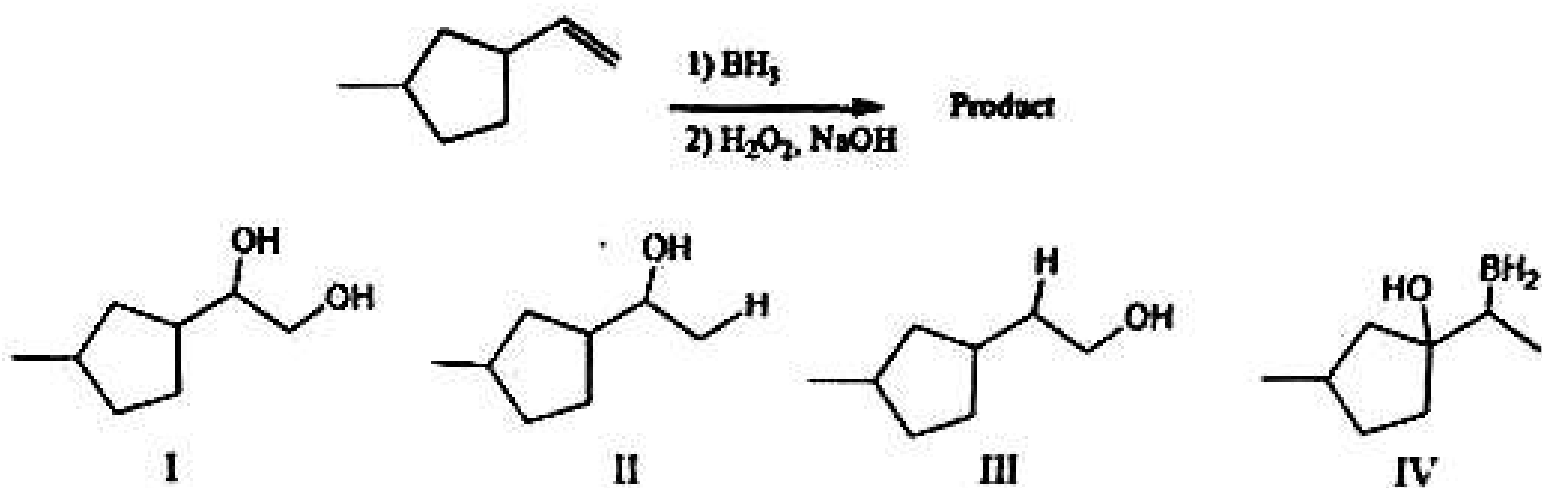
3

- <S> Y
- <C> p-s overlap
- <C+> sp -s overlap
- <C> sp^2 -s overlap
- <C> sp^3 -s overlap

<Q> Which of the following drawings of the direction of the indicated bond polarity is incorrect?



<Q> What is the major product for the reaction shown below?



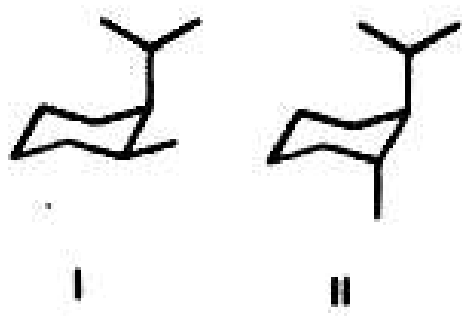
- ☐ Y
☐ I
☐ II
☒ III
☐ IV

<Q> When 3-hexyne is treated with H_2O , H_2SO_4 , and HgSO_4 , the final product is



- ☐ Y
☒ 1
☐ 1 and 4
☐ 3
☐ 2

<Q> Compounds I and II are:



- <S> Y
- <C> Constitutional isomers
- <C> Conformational isomers
- <C+> Cis-trans isomers
- <C> Different compounds (not isomers)

<Q> What is the major product for the reaction shown below?



<Q> The correct order of decreasing boiling point is:



I



II



III



IV

<S> Y

<C+> I>II>IV>III

<C> IV>III>II>I

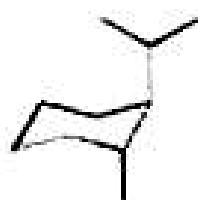
<C> III>II>IV>I

<C> III>IV>II>I

<Q> The most stable conformation of *cis*-1-isopropyl-2-methylcyclohexane is:



I



II



III



IV

<S> Y

<C> I

<C> II

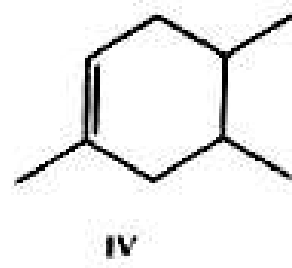
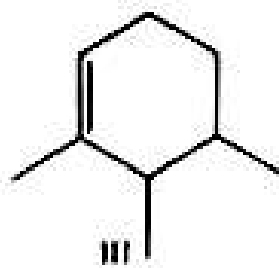
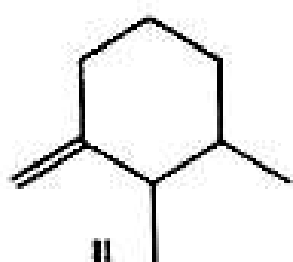
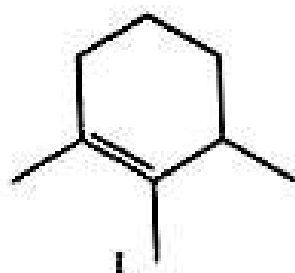
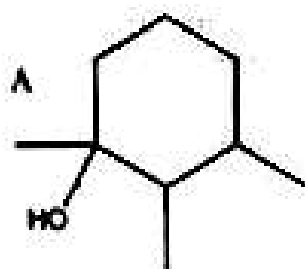
<C> III

<C+> IV

<Q> Compounds I and II are:



<Q> Which of the following alkenes will give alcohol A only when undergo acid-catalyzed hydration?



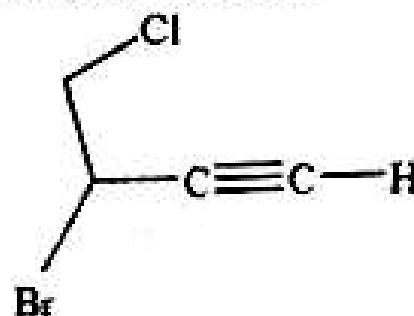
- ☐ Y
- ☐ IV
- ☐ II
- ☐ III
- ☐ II and III

<Q> Cycloalkanes with more than..... carbons in the ring are nonplanar.

- I) 6 II) 5 III) 4 IV) 3

- ☒ Y
☐ I
☐ II
☐ III
☒ IV

<Q> The IUPAC name for the following compound is:



- ☒ Y
☐ 1-chloro-2-bromo-3-butyne
☐ 2-bromo-1-chloro-1-butyne
☒ 3-bromo-4-chloro-1-butyne
☐ 2-bromo-1-chloro-3-butyne

<Q> Which of the following alkenes will give alcohol A only when undergo acid-catalyzed hydration?

